

SYED NAJAM UL HASSAN

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Motivated Software Engineering student (6th Semester) with hands-on experience in AI-based systems, automation workflows, and full-stack development. Skilled in Python, C++, HTML, CSS, and JavaScript with practical experience through projects and internships.

EDUCATION

BS Software Engineering (6th Semester) — Capital University of Science and Technology, Islamabad

FSC Pre-Engineering — Punjab Group of Colleges, Morga Campus

Matriculation (Science) — Quaid-e-Azam Secondary School, Westridge, Rawalpindi

SKILLS & COMPETENCIES

- **Programming Languages:** Python, C++, C, JavaScript
- **Web Development:** HTML5, CSS3, Bootstrap, JavaScript, REST API Development
- **Backend Development:** Flask, Python Backend Development, API Integration
- **AI & Machine Learning:** Pathfinding Algorithms (Dijkstra, A* Search), Natural Language Processing (NLP), Chatbot Development, Virtual Assistant Development, Large Language Models (LLM), Groq API
- **Automation & Workflow:** n8n Workflow Automation, AI Agents, Email Automation, Process Automation
- **Libraries & Frameworks:** OSMnx, NetworkX, Leaflet.js, Airtable, Git, GitHub
- **Software Development:** Object-Oriented Programming (OOP), Full-Stack Development, Data Structures, Algorithms
- **Design & UI/UX:** Figma, Canva, UI/UX Design

PERSONAL PROJECTS

AI Route Planner | Python, Flask, OSMnx, Leaflet.js

- Built a full-stack web application implementing Dijkstra's and A* Search algorithms on real OpenStreetMap road networks, comparing routes by distance, travel time, and nodes explored.
- Developed a Flask REST API backend with OSMnx and NetworkX for road graph caching, geocoding, and routing endpoints.

Jarvis Virtual Assistant | Python

- Developed a voice-controlled virtual assistant using speech recognition and natural language processing (NLP) for task automation.
- Enabled system command execution and query-based responses using Python automation.

AI Chatbot | Python

- Designed a conversational AI chatbot with structured response handling and query-processing logic using NLP techniques.

Signature Verification System

- Developed a signature verification system using pattern recognition and machine learning for identity validation.

AI AUTOMATION (n8n)

Email Agent | n8n, Gmail, Groq

- Built an AI-powered email automation workflow using n8n that triggers on incoming Gmail messages.
- Implemented text classification using Groq LLM to categorize emails and auto-generate context-aware draft replies.

Inventory Agent | n8n, Groq, Airtable

- Built a conversational AI agent using large language models (LLM) to manage product inventory through a chat interface.
- Integrated Airtable database with Groq AI model to search and update inventory records in real time.

VOLUNTEER EXPERIENCE

Volunteer – Junior Jinnah Trust

- Contributed to community service initiatives and social support activities.
- Assisted in organizing and managing volunteer programs.